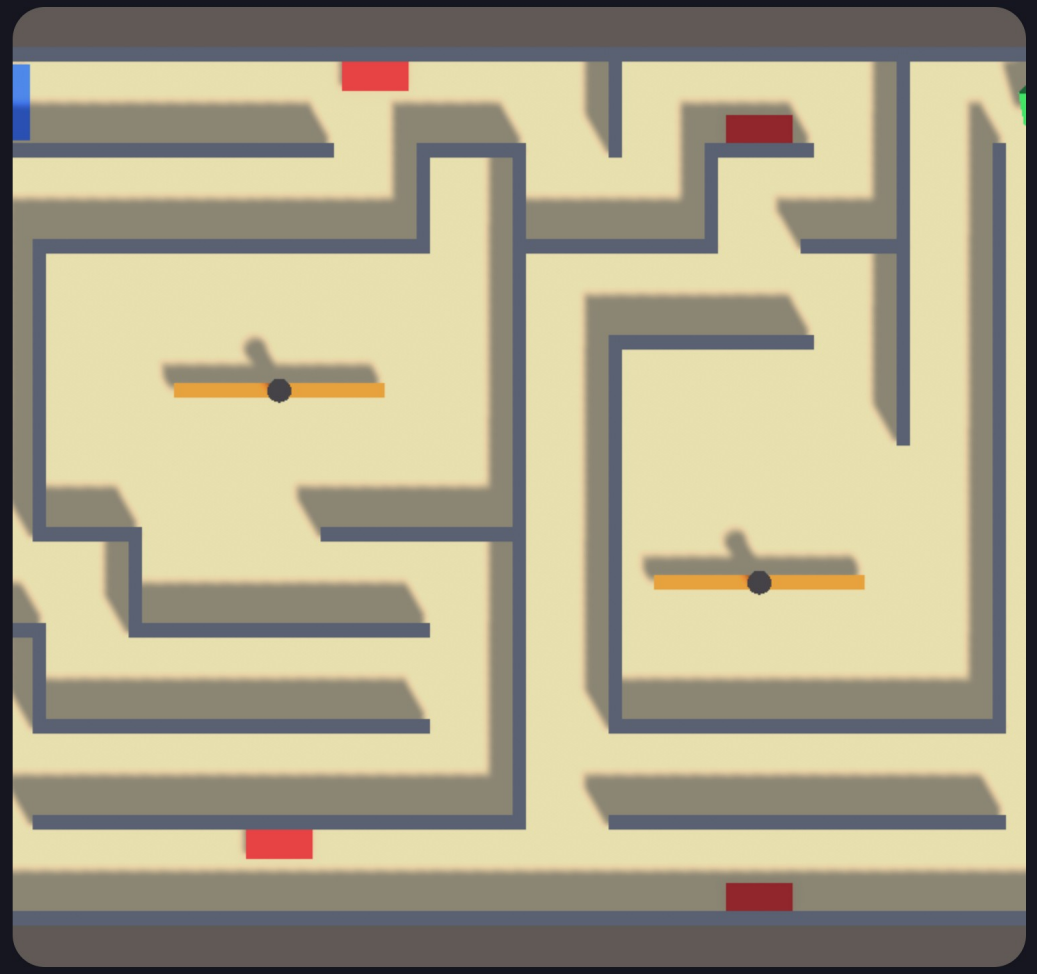
GAME DESIGN MIDTERM

Echo Maze

A maze where every failed run comes back as a ghost, and touching your past self costs a life.

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DMVR504 Game Design and Development · ESLSCA University · Fall 2026



The pitch

Your mistakes don’t disappear. They come back, and they’re in your way.

GENRE

Top-down puzzle-action maze

PLATFORM

PC, keyboard, built in Unity 6

AUDIENCE

12+, fans of short skill puzzles

HOOK

Failed runs replay as deadly ghosts

How to play

Goal

Roll the ball to the green cube. 3 levels.

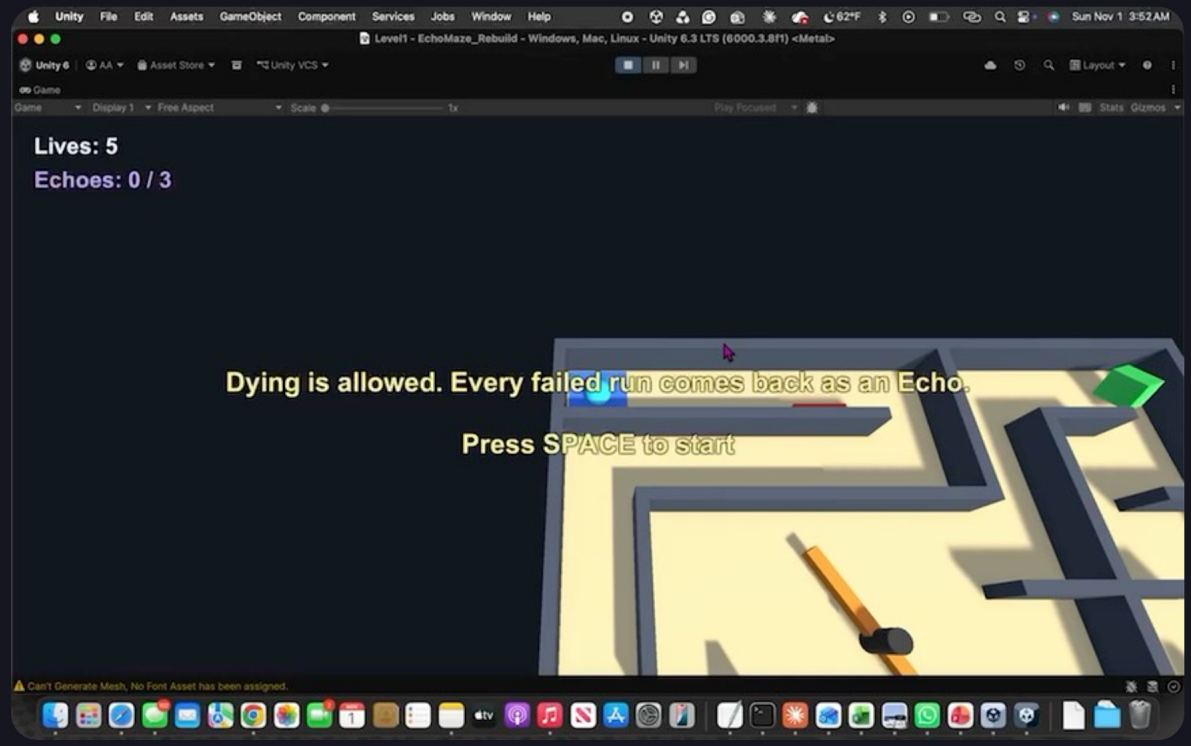
Lose a life

Spikes, spinner arms, or a moving Echo. 5 lives per level.

The Echo

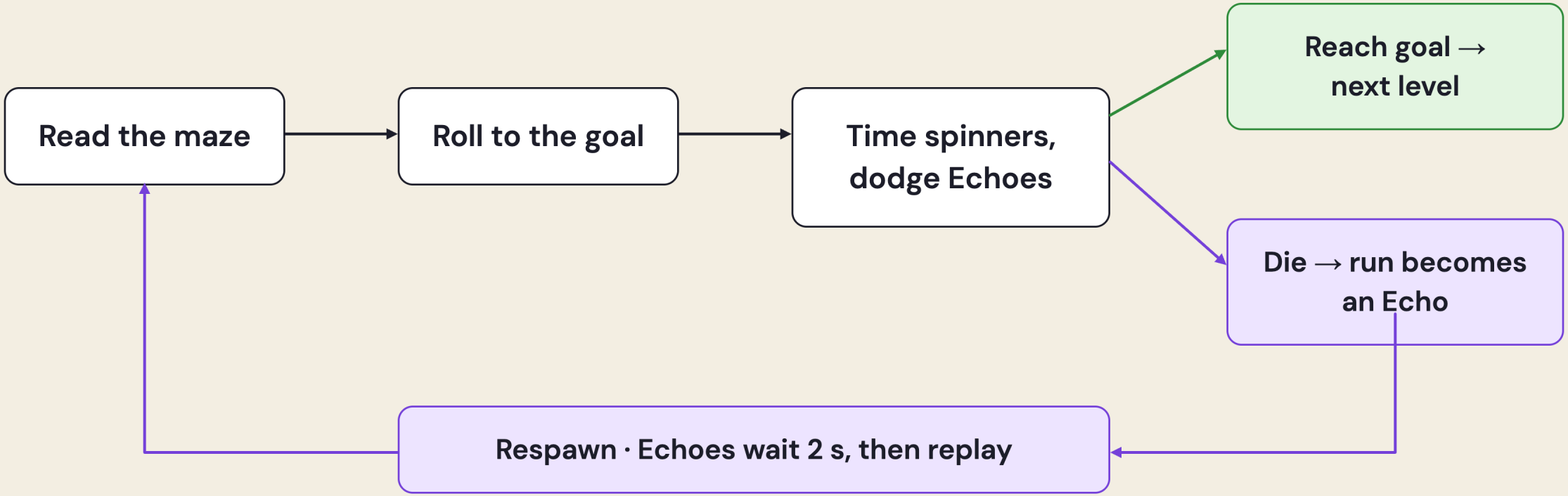
Each death replays as a ghost. It waits 2 s, flickers, then moves. Max 3.

WASD / arrows to roll · SPACE to start · R to restart



Core loop

Every death changes the level, so every retry is a new puzzle.



MDA: rules → behaviour → feeling

Mechanic (we built)	Dynamic (player does)	Aesthetic (player feels)
Each death replays as an Echo; touching it costs a life	Remembers Echo paths, avoids dying on the main route	Challenge, discovery, “chased by myself”
Echo waits 2 s, flickers 0.5 s; max 3	Rushes off the start pad; manages Echo count	Tension, then relief: fair, never hopeless
Ball moves by force (momentum); spinners turn at a fixed speed	Slows before spinner rooms, watches the rhythm, commits	Mastery when you slip through the gap
5 lives, instant restart with R	Retries quickly and experiments	“One more go” motivation

We designed backward (Lecture 3): feeling first, “your mistakes haunt you”, then the behaviour, then the rule.

Five design principles, applied

Clear goals

Intro tip per level; the goal is the only green object

Feedback

Death message names the cause; Echo flickers before moving

Progression

Bigger mazes, more spikes, spinners 50 → 60 → 75°/s

Meaningful choice

Level 3: ≈46 m past 2 spinners or ≈66 m past 3 spike strips

Fairness

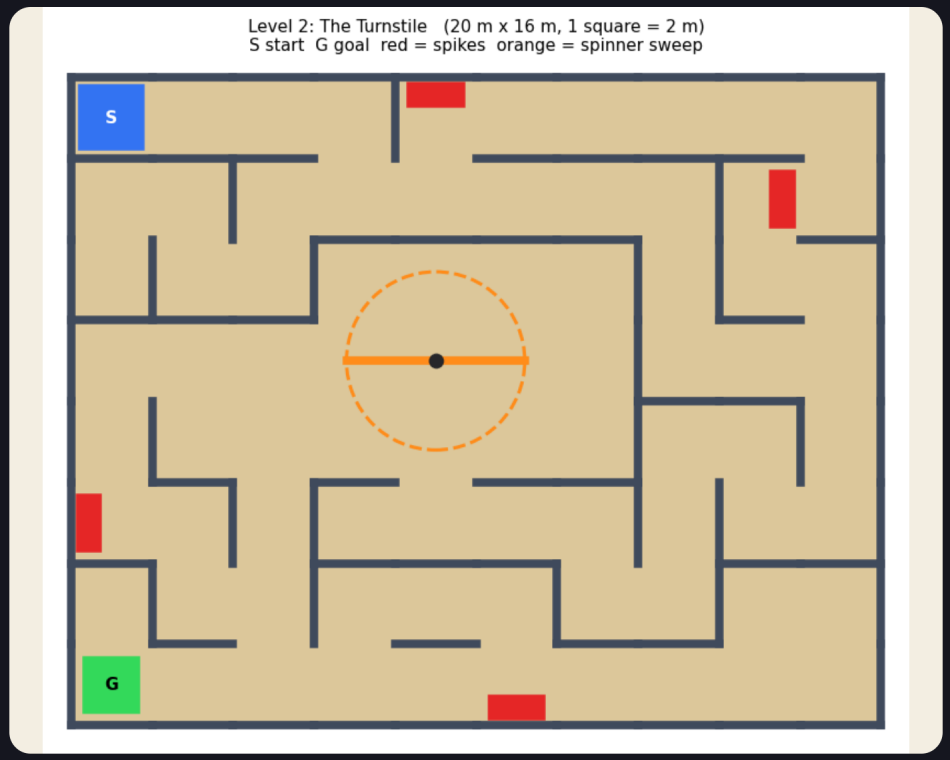
2 s safe window, 3-Echo cap, 5 lives per level

How we test each

Ask testers: goal clear? knew what killed you? which route, why? anything unfair?

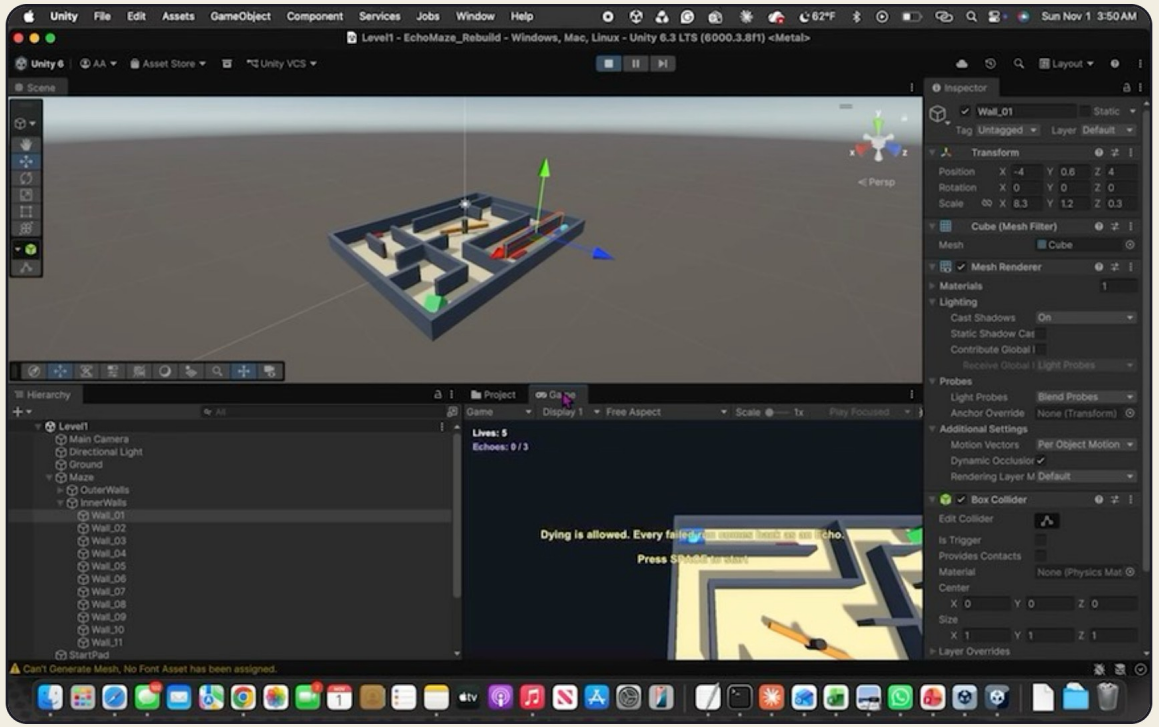
Three levels, rising difficulty

Level	Size	Walls · spikes	Spinners
1 First Echo	16 × 12 m	11 · 3	1 at 50°/s
2 The Turnstile	20 × 16 m	29 · 4	1 at 60°/s
3 Crowded Past	24 × 18 m	25 · 5	2 at 75°/s



Level 2 map: you must cross the spinner room

Built by hand in the editor



Walls = Cubes (L5)

Under **Maze › OuterWalls / InnerWalls**, set by Position and Scale, copied with Duplicate

Hazards = prefabs (L2, L5)

Spikes and Spinner placed under **Hazards**, like the PickUp prefab

Solid vs trigger (L5)

Walls block the ball; spikes, arms, Echoes and the goal are triggers

Five short scripts

Script	From	What it does
PlayerController	Lecture 4	OnMove input, AddForce in FixedUpdate; records the run; OnTriggerEnter by tag
CameraController	Lecture 4	Stores an offset in Start, follows in LateUpdate
Rotator	Lecture 4	transform.Rotate × Time.deltaTime; spinners and the goal
EchoGhost	New	Waits, flickers, then replays the saved path on a loop
GameManager	New	Lives, Echoes, messages, win / lose, loads the next level

The two new scripts use only lecture building blocks: public fields, Start, Update, FixedUpdate, if, and a List.

How an Echo works

1

Record

Every physics step the ball adds its position to a list

2

Die

GameManager makes an Echo prefab copy and hands it the list

3

Wait

2 s hidden with its collider off, then a 0.5 s flicker

4

Replay

One position per step, same speed as you, on a loop

Recorded in FixedUpdate and replayed in FixedUpdate, so the Echo moves exactly like you did (Lecture 4's fixed physics step).

Testing changed the design

38 m = 38 m

Level 2 had no real choice

The safe route was as short as the spinner route. We moved walls so the spinner room is on the only path.

4 / 4

Automated play tests pass

Walls are solid, every hazard costs a life, Echoes replay, the goal wins, Game Over after 5 deaths.

Next: playtests with real players. Questions: goal clear? knew what killed you? which route, why? anything unfair?

Done now, and for the final

Midterm: done

- Movement, walls, follow camera
- Spikes and spinners
- Echo record, replay, delay, cap
- Lives, Game Over, restart
- 3 levels, HUD messages

Final: planned

- Echoes press switches, open doors
- Sound effects and music
- Best time, fewest deaths
- Main menu, 2–3 more levels
- Human playtests and tuning

Questions?

Play it, break it, and tell us where you died.

github.com/seehaman/EchoMaze
Unity 6 · open Assets/Scenes/Level1 · press Play, then SPACE